

Lecture 25

Structure model of 2-to-1 Multiplexer

```
module mux_2_1 (a,s,z);  
input [0:1]a;  
input s;  
output z;  
wire sbar, t1, t2;  
not g1 (sbar, s);  
and g2 (t1, a[0], sbar);  
and g3 (t2, a[1], s);  
or g4 (z, t1, t2);  
endmodule
```

```
module testbench;  
reg [0:1]a;reg s; wire z;  
mux_2_1 uut (.a(a),.s(s),.z(z));  
initial  
begin  
$monitor($time,"a[0]=%b,a[1]=%b  
,s=%b,z=%b",a[0],a[1],s,z);  
#5 a=2'b01;s=0;  
#5 a=2'b01;s=1;  
#5 a=2'b10;s=0;  
#5 a=2'b10;s=1;  
#5 $finish;  
end  
endmodule
```

